



General Sir John Kotelawala Defence University

Faculty of Computing

Department of Computer Engineering

COE12012_Fundamentals of Electrical Engineering

Activity Instructions

1. Students must **upload the completed activity** within lecture hours.
Late submissions will **NOT** be accepted under any circumstance.
2. The assignment must be submitted **as a single PDF file**.
3. Rename your submission using the following format:
DBCOE_26_XXXX.pdf
(XXXX – Your Index Number)
4. Students must upload the screenshots for all proteus tasks.
5. Please verify your file before uploading.
6. Students are required to provide **your own descriptive analytical answers**.

**** Submission after the deadline or incorrect file won't be accepted.**

Activity 04

1. OBJECTIVES

- Investigate the transient response (charging and discharging time constants) of an RC circuit in Proteus.
- Observe waveform shaping using RC Integrator and Differentiator circuits.

2. REQUIRED COMPONENTS & VIRTUAL INSTRUMENTS (PROTEUS)

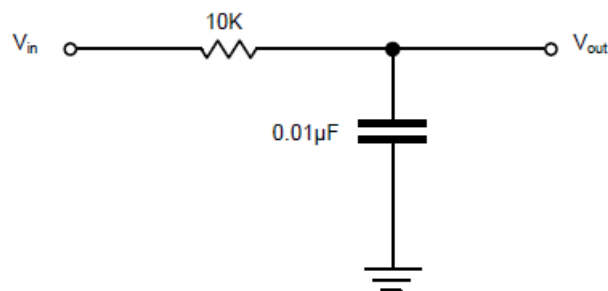
Search and import the following components from the Proteus Pick Devices (P) library:

Component / Instrument	Proteus Library Keyword / Path	Quantity
Capacitors	Cap	As needed
Resistors	RES (Interactive Resistor)	Assorted
Signal Generator	Generator Mode	1
Frequency Response Graph	Graph Mode-> Frequency	1
Oscilloscope	Virtual Instruments Mode -> Oscilloscope	1

3. EXPERIMENTAL PROCEDURES & TASKS

Task A: RC circuit.

A.1 Circuit— RC circuit.

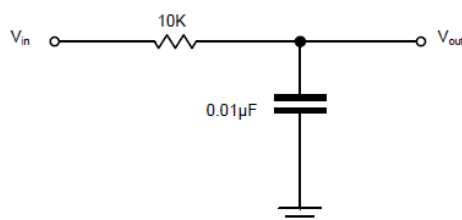


A.2 Procedure

1. Build a series RC circuit in Proteus using $R = 10\text{ k}\Omega$ and $C = 0.01\text{ }\mu\text{F}$ connected to Ground.
2. Set the Signal Generator to produce a Square Wave at 5 V_{p-p} , 500 Hz frequency.
3. A.3 Connect Channel A of the Oscilloscope across the generator (V_{in}) and Channel B across the capacitor (V_{out}).

Task B: Low-Pass & High-Pass Filter Circuits

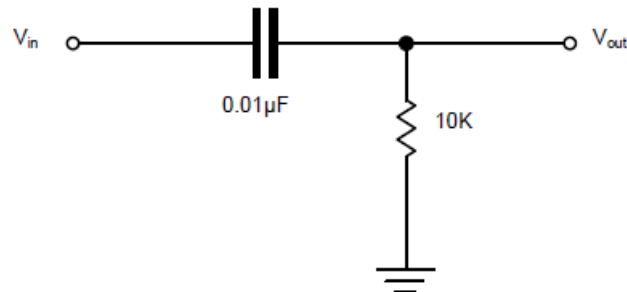
B.1 Low-Pass Circuit



B.2 Procedure

1. Construct a Low-Pass Filter using $R = 10\text{ k}\Omega$ in series with input and $C = 0.01\text{ }\mu\text{F}$ in parallel with output to Ground.
2. Set $V_{in} = 1\text{ Vp-p}$ (Sine wave). Drive the circuit starting at 100 Hz.
3. Measure V_{out} and compute Voltage Gain $G_v = V_{out} / V_{in}$.

B.2 High-Pass Circuit

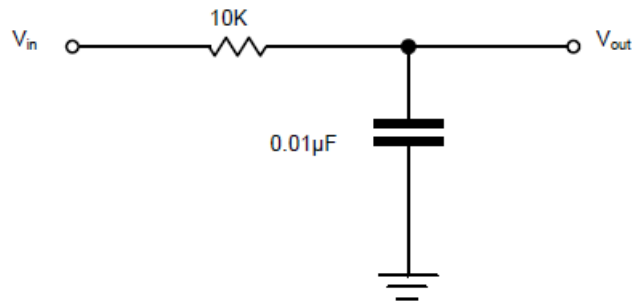


B.3 Procedure

1. Swap components to build a High-Pass Filter: $C = 0.01\text{ }\mu\text{F}$ in series with input and $R = 10\text{ k}\Omega$ connected to Ground.
2. Set $V_{in} = 1\text{ Vp-p}$ (Sine wave). Drive the circuit starting at 500 Hz.

Task C: Integrator Circuit

C.1 Circuit

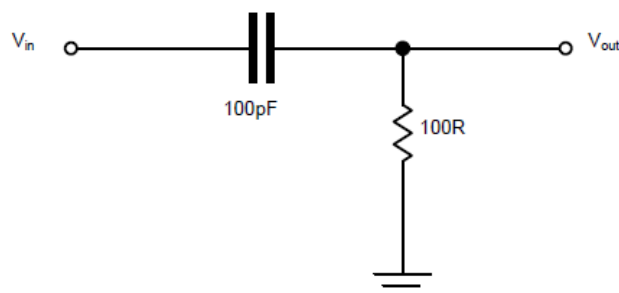


C.2 Procedure

1. Integrator Circuit: Connect $R = 10\text{ k}\Omega$ and $C = 0.01\text{ }\mu\text{F}$. Drive with 100 kHz input signals.
2. Observe V_{out} waveforms on Oscilloscope for (a) Square Wave, (b) Triangle Wave, (c) Sine Wave inputs.

Task D: Differentiator Circuit

D.1 Circuit



D.2 Procedure

1. Differentiator Circuit: Rebuild circuit using $C = 100 \text{ pF}$ and $R = 100 \text{ }\Omega$. Drive with 100 kHz input signals.
2. Observe V_{out} waveforms on Oscilloscope for (a) Square Wave, (b) Triangle Wave, (c) Sine Wave inputs.